

Does an AI One-Sentence Summary Improve Museum Label Understanding?

A simple within-subjects experiment with 20 XJTU students or staff.

Members: Member 1 · Member 2 · Member 3

Evaluation Methods and Statistics

Study in progress



SYSTEM

XJTU Museum labels

Original label compared with one AI-generated summary sentence.

CONTEXT

Exhibit interpretation

University history, cultural exchange, visual art, and traditional culture.

PARTICIPANTS

Campus visitors

20 students or staff; one 10-minute online session.

METHOD

Within-subjects experiment

Four matched exhibit themes, two label formats, counterbalanced order.

01 PROBLEM & MOTIVATION

Museum labels can be dense for a linguistically diverse campus audience.

Gap: no local evidence shows whether AI rewrites improve comprehension without reducing trust.

02 QUESTIONS & HYPOTHESES

RQ1: Does AI summary improve comprehension?

RQ2: Does it make the label feel easier?

H1: Higher comprehension accuracy.

H2: Higher perceived ease.

03 EXPERIMENT DESIGN

Each participant reads three original and three AI-summary labels in counterbalanced order.

1

Consent

Purpose + rights

2

Practice

One example

3

Read

Six labels

4

Answer

1 question each

5

Rate

Ease 1-7

04 MEASURES & SAMPLE

Comprehension

1 question per label

Perceived ease

1-7 rating

Reading time

Secondary descriptive measure

Sample: 20 XJTU students or staff; paired data for both conditions.

05 PRINCIPAL FINDINGS

H1

Comprehension difference

H2

Ease difference

Support

Reading-time description

Pending data: insert real figures and anonymised evidence after data collection.

06 LIMITS & RECOMMENDATIONS

Limit

Three tests and 20 participants.

Review

Curator checks AI summaries.

Keep original

Do not replace full label.

Disclose

Mark AI assistance.

07 REFERENCES

XJTU Museum, official information, 2024.

J. Sweller et al., Cognitive Load Theory, 2011.

S. Amershi et al., "Guidelines for human-AI interaction," CHI, 2019.

B. Serrell, Exhibit Labels, 2015.

Acknowledgement: Experiment inspired by the public exhibition themes and communication context of XJTU Museum. No ownership of Museum collections or original texts is claimed.